

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M. Tech.)

Semester – II University Examination May - 2011

M. Tech. (CAD/CAM)

ME – 710 Industrial Automation and Robotics

Date: 19.05.2011, Thursday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

- Q1 (a) Differentiate between Fixed Automation & Programmable Automation. 04
- (b) What are the advantages of using Automated Guided Vehicle System (AGVS) in Flexible Manufacturing System (FMS). 04
- (c) Draw Hydraulic symbols for: 04
- i) Single acting cylinder
 - ii) Intensifier
 - iii) 2/2 directional valve
 - iv) Pressure reservoir

OR

- Q1 (a) Explain briefly various types of Transfer machines. 04
- (b) List and explain various types of feeders used in material handling. 04
- (c) What is Fluidics? Explain its advantages and limitations 04
- Q2 (a) What is ASRS? What are its types? 05
- (b) What design information is required to a designer for designing hydraulic circuit? 06

OR

- Q2 (a) Write short note on Coanda Effect. 05
- (b) Classify material handling systems on the basis of main features, advantages and limitations. 06

- Q3 (a) Write short notes on *any three*: 12
- (i) Classification of fluidic devices
 - (ii) Band and Belt conveyors
 - (iii) Flexible Automation
 - (iv) OR-NOR Gate
 - (v) Fluidic Sensors

SECTION – II

- Q4** Write short notes on *any three*: 12
- (i) Adaptive control
 - (ii) Type of robot controllers
 - (iii) Manipulator Transformation Matrix
 - (iv) Newton Euler dynamic modeling of manipulator
 - (v) D'Alembert's Principle
 - (vi) SCARA

- Q5 (a)** With neat sketch explain controller architecture and functions of elements for a robot system. 04
- (b)** Describe tactile sensors for robotic applications. 04
- (c)** Describe fundamentals of image processing for robotic vision system. 04

OR

- Q5 (a)** Compare pneumatic and hydraulic drives of robot. 04
- (b)** Discuss how resolution of a robotic manipulator is affected by mechanical inaccuracies. 04
- (c)** One of the axes of a robot is a telescoping arm with a total range of 0.75 m. The robot's control memory has a 32-bit storage capacity for this axis. Determine the control resolution for the axis. 04

- Q6 (a)** Explain the joint notation scheme for various configurations of robot manipulator. 04
- (b)** Write down following for a PUMA 600 robotic manipulator 07
- (i) Structure or configuration
 - (ii) Schematic diagram and joint co-ordinate system
 - (iii) Table of joint parameter
 - (iv) Overall direct kinematics transformation matrix

OR

- Q6 (a)** Explain various representation schemes for robot kinematic chain. 04
- (b)** Write down following for a Rhino-XR2 robotic manipulator 07
- (i) Structure or configuration
 - (ii) Schematic diagram and joint co-ordinate system
 - (iii) Table of joint parameter
 - (iv) Overall direct kinematics transformation matrix
